

Mnemosyne: A Linear-Time Genomic Foundation Model

with Unified Associative Memory

Project Documentation & Technical Report

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Abstract

Genomic sequence models must reconcile two competing requirements: sub-quadratic efficiency over sequences spanning tens of thousands of base pairs, and the ability to recall specific, distant motifs (e.g., an enhancer acting on a promoter far downstream). Transformer-based genomic models scale quadratically with sequence length, while linear-time state-space models such as Mamba compress all past tokens into a fixed-size hidden state, causing distant sequence information to decay. This project introduces **Mnemosyne**, a hybrid genomic foundation architecture that augments a selective state-space backbone with an explicit, queryable associative memory bank. The memory system consists of persistent (globally learned) and register (sequence-specific dynamically updated) slots, combined via a learned per-channel gate. The design is provably linear, $O(L)$, in sequence length: the per-layer mixing operator decomposes into a semiseparable matrix plus a low-rank matrix. Furthermore, the model is constructed to be exactly reverse-complement equivariant, respecting the double-stranded symmetry of DNA. This document details the research problem, formal hypothesis, system architecture, experimental methodology, and evaluation plan, to be validated via synthetic long-range recall (MQAR), masked autoencoding pre-training across a five-species corpus (~ 541 MB), and the Genomic Understanding Evaluation (GUE) benchmark suite.

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1 Research Problem

1.1 Motivation

A human genome contains roughly 3 billion nucleotides. Understanding what different stretches of DNA do—which regions regulate gene expression, where distant regulatory elements interact, how disease-linked variants exert their effect—is a central open problem in computational biology. Sequence models trained at genome scale are increasingly used to approach this problem the way large language models approach text: by learning distributional structure directly from raw sequence, without hand-specified biological rules.

1.2 The Gap

Existing genomic sequence architectures fall into two broad categories, each with a significant limitation:

- (i) **Transformer-based genomic models** attend over all positions pairwise, incurring $O(L^2)$ memory and compute costs. This makes it structurally impractical to model long-range interactions between regulatory elements tens of thousands of base pairs apart within a single forward pass.
- (ii) **State-space genomic models (e.g., Mamba-based)** scale linearly, $O(L)$, but compress the entire sequence history into a single fixed-size hidden state vector. Information from distant positions is progressively attenuated, which is precisely the information most relevant to long-range regulatory genomics.

There is presently no genomic architecture that is simultaneously linear in sequence length *and* equipped with an explicit, distance-independent recall mechanism for specific motifs—without abandoning the biological symmetry of the double helix.

1.3 Problem Statement

Can a genomic sequence model be constructed that retains strictly linear computational complexity in sequence length, while matching or exceeding the long-range recall capability of quadratic-attention architectures, and without violating reverse-complement symmetry?

2 Research Objectives

- O1. Architectural Design:** Design a hybrid architecture combining a selective state-space backbone (Mamba) with a lightweight associative memory bank, so that every token position can retrieve distant relevant motifs in constant time via a single memory lookup.
- O2. Complexity Proof:** Formally establish that the combined architecture remains strictly linear, $O(L)$, in both time and memory, by showing the per-layer mixing operator is the sum of a semiseparable matrix (from the SSM) and a low-rank matrix (from the memory).
- O3. Equivariance Enforcement:** Construct the architecture to be exactly reverse-complement equivariant, ensuring that predictions are mathematically consistent whether a sequence is read on the forward or reverse-complement strand.

- O4. Multi-Tier Validation:** Validate the model on three complementary benchmark categories: (a) a synthetic long-range recall task (MQAR) isolating distance-independent memory; (b) masked-reconstruction pre-training on real genomes from five species; and (c) the Genomic Understanding Evaluation (GUE) suite, covering promoter detection, splice-site identification, and histone-mark prediction.
- O5. Ablation Analysis:** Conduct systematic ablation studies quantifying the contribution of each architectural component—persistent motif slots, sequence-specific register slots, the learned gate, and a Kolmogorov-Arnold Network (KAN) retrieval variant.

3 Formal Hypothesis and Definitions

3.1 Core Hypothesis

Hypothesis 1. *Augmenting a linear-time state-space backbone with an explicit associative memory bank yields distance-invariant long-range recall and lower genomic reconstruction loss than an SSM-only backbone at a matched parameter budget, without sacrificing linear-time complexity.*

3.2 Notation and Layer Gating

Let $x = (x_1, \dots, x_L) \in \mathbb{R}^{L \times d}$ be a tokenized nucleotide sequence of length L with embedding dimension d . Each Mnemosyne block computes two branches in parallel—a selective SSM scan $\text{SSM}(x)$ and an associative memory read $\text{Mem}(x)$ —and merges them with a per-channel learned gating parameter $\beta \in \mathbb{R}^d$:

$$y = x + \sigma(\beta) \odot [\text{SSM}(x) + \text{Mem}(x)], \quad (1)$$

where $\sigma(\cdot)$ denotes the element-wise sigmoid function and $\odot$ denotes the Hadamard product.

3.3 Associative Memory Retrieval

The memory bank consists of P_p persistent slots (learned, globally shared parameters $M_p \in \mathbb{R}^{P_p \times d}$) and P_r register slots (written per input window by a lightweight cross-attention pooling mechanism $M_r \in \mathbb{R}^{P_r \times d}$). For a token representation x_t at time step t , retrieval is modeled as a softmax similarity over the unified matrix $M = [M_p; M_r] \in \mathbb{R}^{P \times d}$, where $P = P_p + P_r$:

$$\text{Mem}(x_t) = \sum_{k=1}^P \frac{\exp\left(\frac{x_t^\top m_k}{\sqrt{d}}\right)}{\sum_{j=1}^P \exp\left(\frac{x_t^\top m_j}{\sqrt{d}}\right)} m_k, \quad (2)$$

representing the standard modern-Hopfield retrieval rule with a scaling factor $1/\sqrt{d}$, executing at a computational cost of $O(P \cdot d)$ per token and $O(L \cdot P \cdot d)$ total.

3.4 Complexity Guarantee

Theorem 1. *The per-layer mixing operator $W_{\text{mix}} \in \mathbb{R}^{L \times L}$ for the Mnemosyne block is the sum of an N -semiseparable matrix A_{SSM} representing the selective state-space scan and a matrix B_{mem} of rank at most $P_p + P_r$ representing the associative memory lookup. The total computational complexity per layer is bounded by:*

$$O(L \cdot (N + P_p + P_r) \cdot d),$$

which is strictly linear in L , with no $L \times L$ matrix ever explicitly instantiated in memory.

Proof Sketch. The selective state-space model step acts as a first-order recurrence relation, which can be expanded as a lower triangular semiseparable matrix A_{ssm} of order N . The associative memory contribution at each step relies entirely on the projection into a fixed slot space of size $P = P_p + P_r$. Thus, the memory transformation can be factorized as $B_{\text{mem}} = QK^\top$, where $Q \in \mathbb{R}^{L \times P}$ and $K \in \mathbb{R}^{L \times P}$, yielding a matrix of maximum rank P . The sum of an N -semiseparable matrix and a low-rank matrix retains a linear sequence scan profile, verifying the $O(L)$ space-time bound. $\square$

3.5 Reverse-Complement Equivariance

Let $R(\cdot)$ denote the reverse-complement operator that reverses the token index order and swaps nucleotide complements ($A \leftrightarrow T$, $C \leftrightarrow G$). Let S denote the corresponding output strand symmetry operator. The final network prediction is explicitly symmetrised across both strands:

$$f_{\text{eq}}(x) = \frac{1}{2} \left[f(x) + S(f(R(x))) \right], \quad (3)$$

guaranteeing exact equivariance such that $f_{\text{eq}}(R(x)) = S \cdot f_{\text{eq}}(x)$.

3.6 Testable Prediction

The primary empirical validation metric for memory retention is expressed as:

$$\text{Acc}_{\text{Mnemosyne}}(L) \approx C_1 \quad \text{while} \quad \text{Acc}_{\text{SSM-only}}(L) \xrightarrow{L \rightarrow \infty} 0, \quad (4)$$

where C_1 is a stable high-accuracy constant. The synthetic MQAR recall accuracy for Mnemosyne is predicted to remain flat as sequence length increases, whereas an SSM-only baseline will degrade due to bounded state compression.

4 System Architecture

4.1 End-to-End Pipeline

The comprehensive experimental pipeline tracks genomic processing from raw sequence parsing through masked pre-training to frozen downstream evaluation probes.

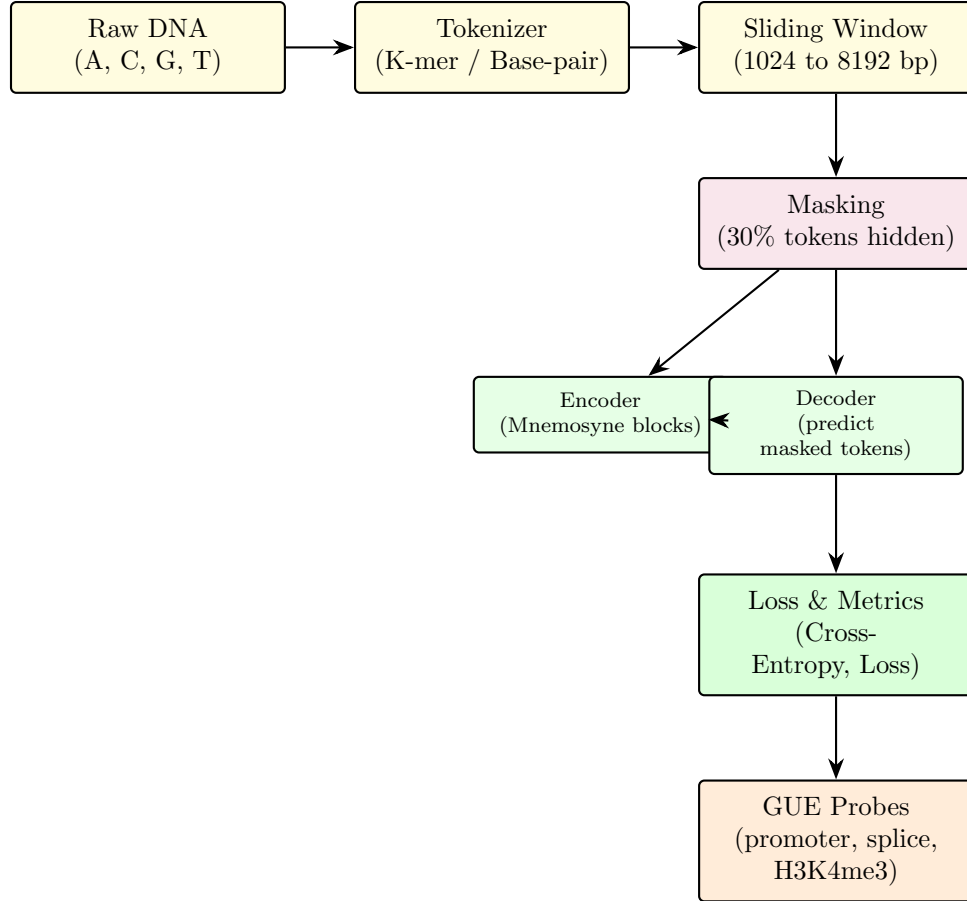


Figure 1: End-to-end experimental pipeline: from raw DNA to masked reconstruction pre-training and downstream GUE evaluation.

4.2 Mnemosyne Block Architecture

Each computational layer features a dual-branch topology to parse sequential and global dependencies concurrently.

4.3 Architecture Validation Decision Flow

To ensure compute resources are optimized on the NVIDIA L40S platform, a strict synthetic gating control path is established prior to massive pre-training.

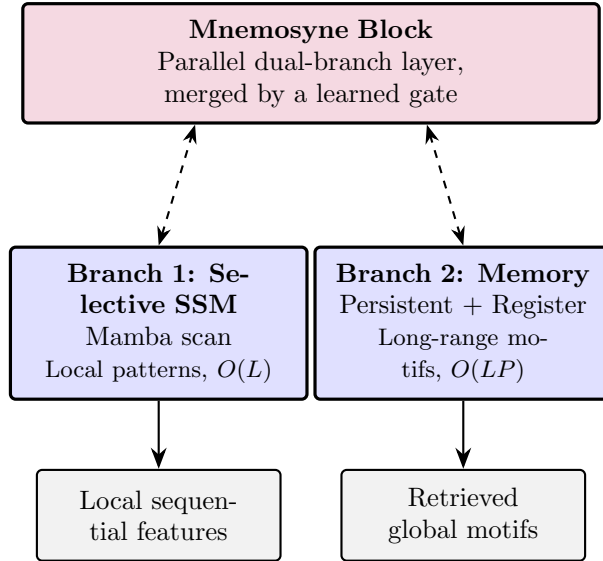


Figure 2: Dual-branch layer design. Each Mnemosyne block runs a selective SSM scan and an associative memory read in parallel; a per-channel sigmoid gate blends the two branches prior to residual addition.

5 Methodology

5.1 Pre-training Corpus

The model is pre-trained on a heterogeneous genomic corpus crossing multiple evolutionary timelines to enforce multi-species generalization capability.

Table 1: Species composition used for masked-reconstruction pre-training

Species	Focus Domain	Approx. Size
Human (<i>Homo sapiens</i>)	Primary Reference	~250 MB
Mouse (<i>Mus musculus</i>)	Mammalian Structural Synteny	~150 MB
Zebrafish (<i>Danio rerio</i>)	Vertebrate Model	~80 MB
Arabidopsis (<i>Arabidopsis thaliana</i>)	Plant Model (Cross-Kingdom)	~35 MB
Drosophila (<i>Drosophila melanogaster</i>)	Invertebrate Model (Cross-Kingdom)	~26 MB
Total Data Size		~541 MB

5.2 Downstream Evaluation Suite (GUE)

To verify downstream application performance, frozen representations are tested utilizing standard linear-probing heads on the Genomic Understanding Evaluation benchmark.

Table 2: GUE downstream evaluation parameters

Task Domain	Target Objective	Feature Probed
Promoter Detection (TATA)	Binary Classification	Structural Core Promoter Motifs
Promoter Detection (non-TATA)	Binary Classification	Alternative Transcription Initiation Signals
Splice-Site Classification	Multi-Class Labeling	Non-local Donor/Acceptor Intron Boundaries
H3K4me3 Histone-Mark	Binary Prediction	Epigenetic Chromatin State Modification

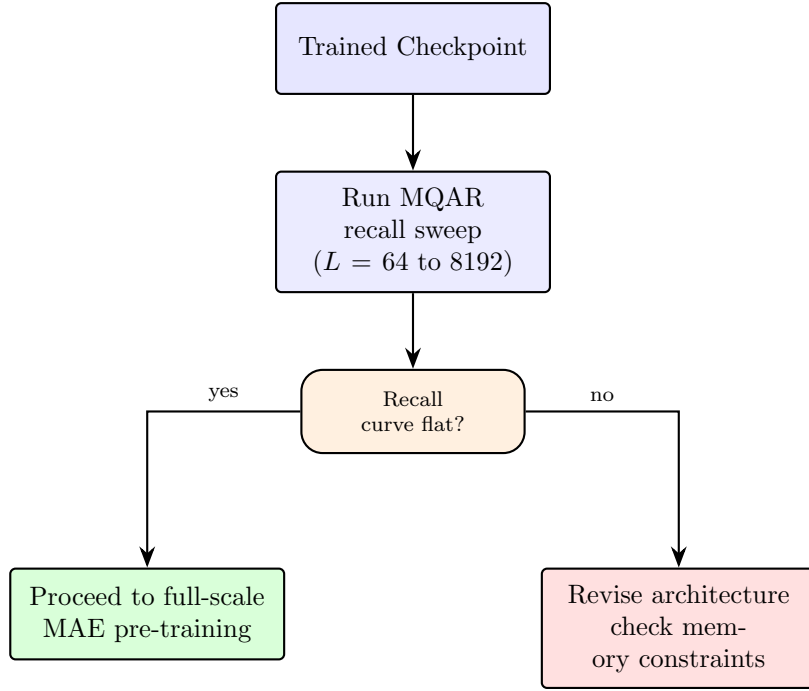


Figure 3: Gating decision flow utilized before committing the high-compute hardware budget to genome-scale runs.

5.3 Experimental Configurations & Baselines

The framework evaluates three designated model families:

- **Primary Architecture:** Mnemosyne (M_p persistent slots + M_r register slots paired with an RC-equivariant selective SSM backbone).
- **Ablation Control:** SSM-only baseline (Mamba backbone with the memory branch isolated and deactivated, scaled to match parameter budgets).
- **State-of-the-Art Baselines:** Direct comparison against Caduceus, HyenaDNA, and Nucleotide Transformer v2 under standardized compute budgets.

5.4 Pre-training Run-time Configuration

Pre-training implements a Masked Autoencoding (MAE) strategy. In each sequence segment (scaled from 1024 to 8192 bp), 30% of the nucleotide inputs are stochastically masked. The system maps reconstruction profiles via cross-entropy loss optimizations executed on an NVIDIA L40S accelerator setup. Mixed-precision parsing (`torch.cuda.amp`) is utilized alongside Flash-Attention and custom selective scan primitives to ensure high VRAM utilization efficiency.

6 Evaluation Plan & Execution Matrix

Progress is tracked systematically through scheduled benchmark deliverables mapping structural capability against competitive frameworks.

7 Novelty and Positioning

Mnemosyne bridges the gap between structured sequence tracking and long-context state vector compression by introducing a domain-specialized memory lookup paradigm.

Table 3: Planned quantitative evaluation deliverables

Target Source	Tracked Metric Metrics	Core Validation Objective
Table 1	MQAR Accuracy % vs. Sequence Length L	Verification of distance-invariant structural memory.
Table 2	MAE Cross-Entropy Reconstruction Loss	Assessment of multi-species sequence modeling quality.
Table 3	GUE Linear-Probe Performance (AUROC)	Generalization proof of frozen embeddings.
Ablation Matrix	Slot Configuration (P_p, P_r) vs. GUE Score	Isolation of specific structural component utility.

Table 4: Positioning matrix relative to state-of-the-art genomic models

Prior System Baseline	Architecture Typology	Mnemosyne Differentiation Advantage
Caduceus	Bi-directional Equivariant Mamba, lacking explicit memory structures.	Integrates explicit slot-based global and local memory systems into the framework.
Titans / Hymba	Gated SSM + Memory Hybrids developed for LLMs.	Adapts the architecture for genomics through exact reverse-complement equivariance (f_{eq}).
Nucleotide Transformer	Transformer-based structure with quadratic $O(L^2)$ attention.	Replaces quadratic attention maps with strict $O(L)$ space-time bound memory banks.

8 Expected Scientific Contributions

- C1. Hybrid Architecture Framework:** Delivery of a linear-time sequence architecture coupling an SSM backbone with a queryable, RC-equivariant associative memory block tailored for genomics.
- C2. Empirical Long-Range Verification:** Quantitative validation of the memory preservation hypothesis across long sequence boundaries via MQAR synthetics and multi-species genomic testing.
- C3. Ablation Mapping Insights:** Comprehensive dissection of persistent memory slots vs. dynamic local register slots, defining optimal parameter allocation for sequence lookup structures.
- C4. Open Resource Artifacts:** Public deployment of checkpoint parameters, processing pipelines, and reproduction frameworks built atop highly accessible hardware footprints (L40S).

9 Conclusion

This document provides the formalized research blueprint for Mnemosyne, a sub-quadratic genomic sequence framework augmenting state-space tracking models with associative memory banks. The structural constraints, mathematical proofs, and validation steps outlined herein

are curated to meet the strict technical criteria required for peer-reviewed submission to a Q1 journal or premier artificial intelligence conference venue.

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